Comparisons and uncertainty

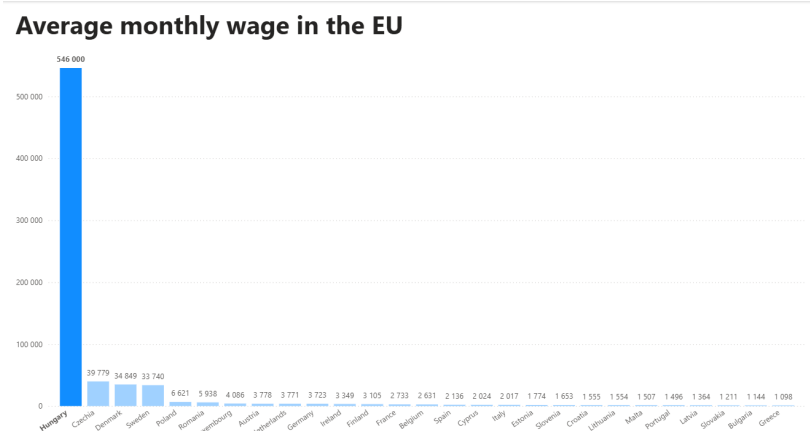
Data Visualization and Storytelling

Lecture 2 11 November 2026

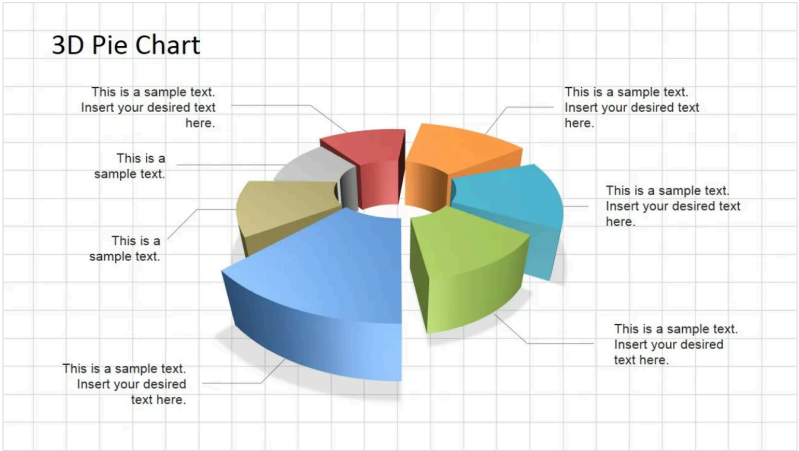
Ulrich Wohak

Your collected figures, your redesigns, then distributions and uncertainty.

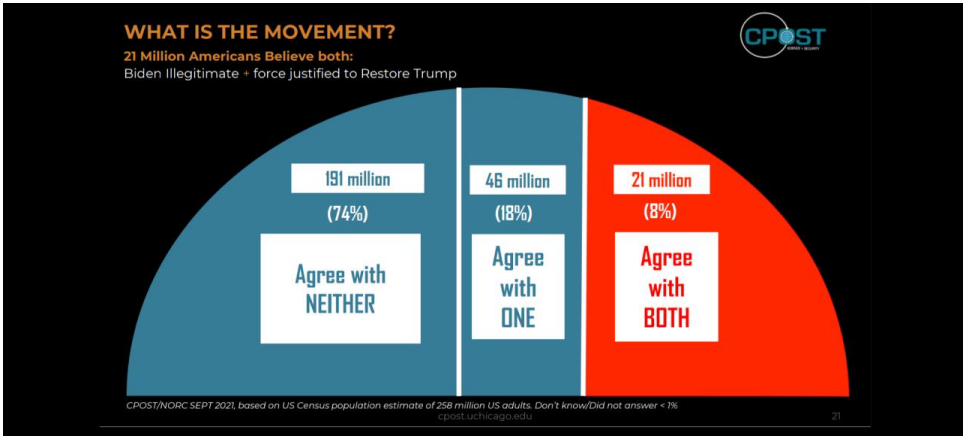
Average monthly wages in the EU



A 3D pie-chart template



Survey shares in a semicircle



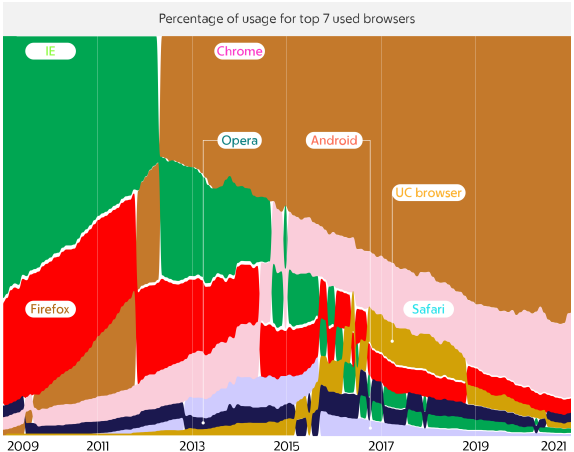
Cigarette prices



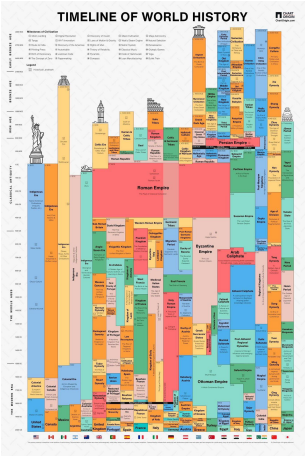
Weekly initial unemployment claims



Browser use over time

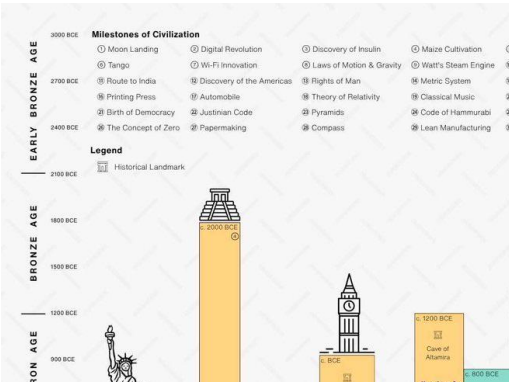


A timeline of world history

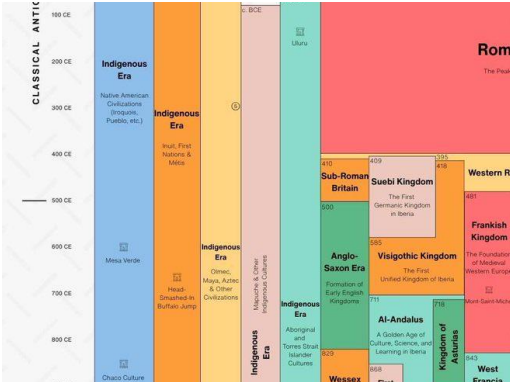


Reading the timeline's scale

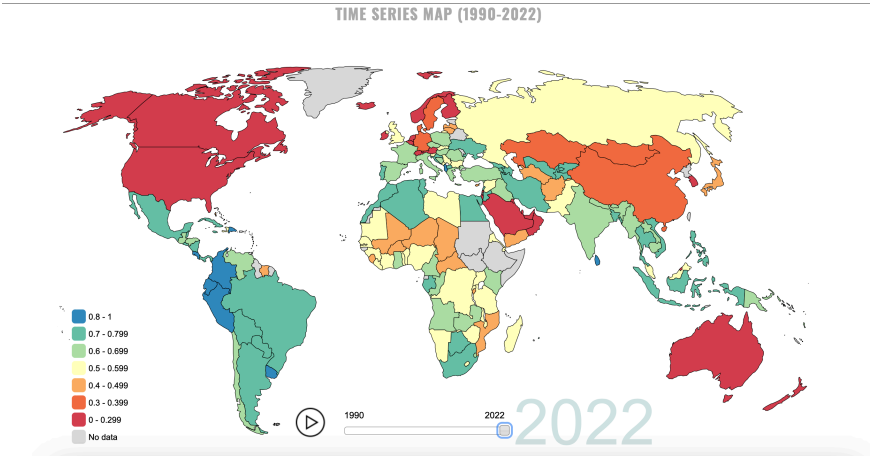
Earlier periods



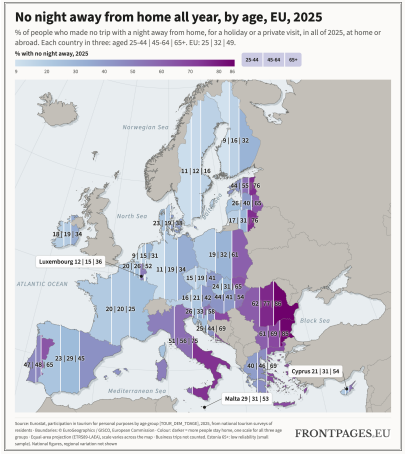
Later periods



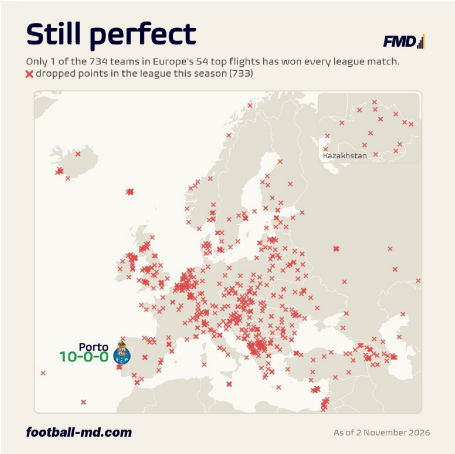
An animated map in a screenshot



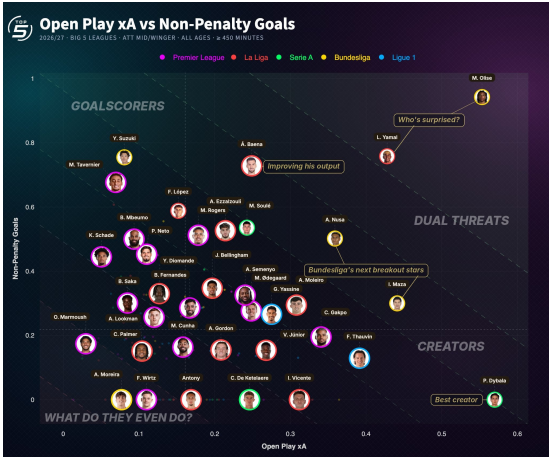
Tourism by country and age



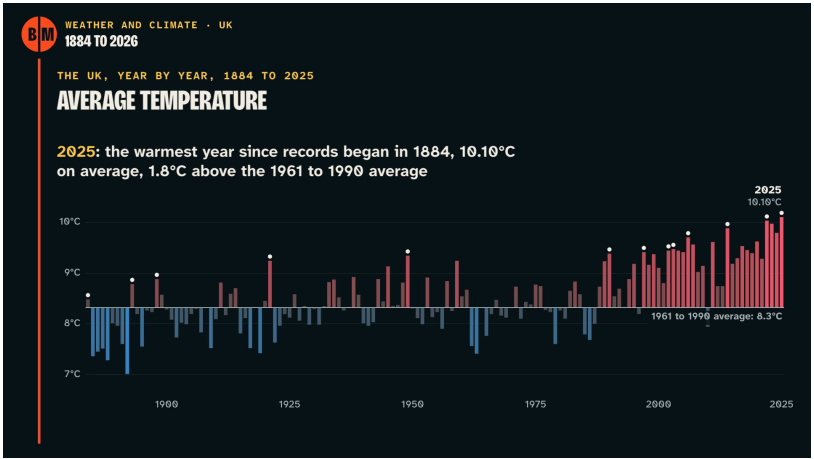
One team highlighted on a map



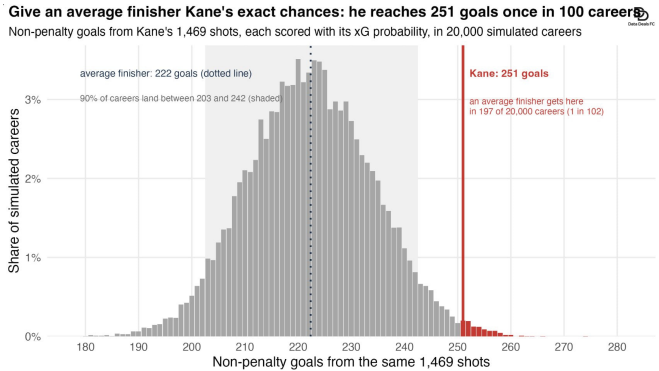
Football players on two dimensions



Temperature relative to a baseline



Kane and simulated careers

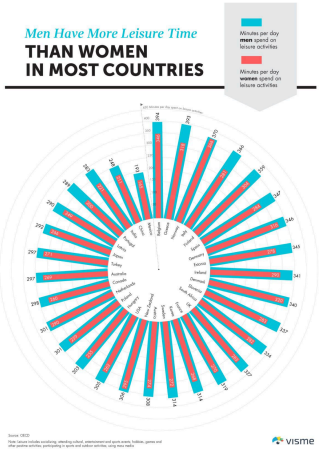


How to read: each bar is the share of 20,000 simulated careers that ended on that many goals. In every simulation an average finisher takes Kane's exact 1,469 non-penalty league shots and each shot goes in with its Understat xG as the probability, so the distribution is what the chances alone produce. It centres on 222 goals (the xG total) with a spread of about 12; ninety per cent of careers land between 203 and 242. Kane scored 251. Only 197 of the 20,000 careers reached that many: about 1 in 102. Add the penalties (62 attempts at Understat's 0.76 each) and the same test on all 308 goals gives 17 in 20,000, about 1 in 1,250. The most any simulated career managed was 274.

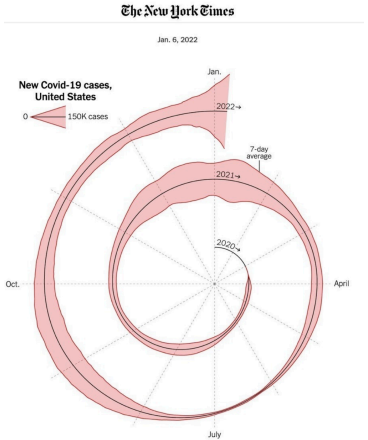
Data: Understat (top five leagues, 2014/15-2025/26) | Shown: Harry Kane's non-penalty league shots, Premier League 2014/15-2022/23 and Bundesliga 2023/24-2025/26; simulation seeded, 20,000 draws

Created by: Martin Angelov @ DataDealsFC.com

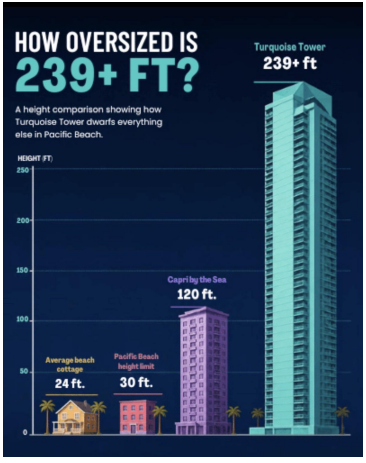
Leisure time by country and gender



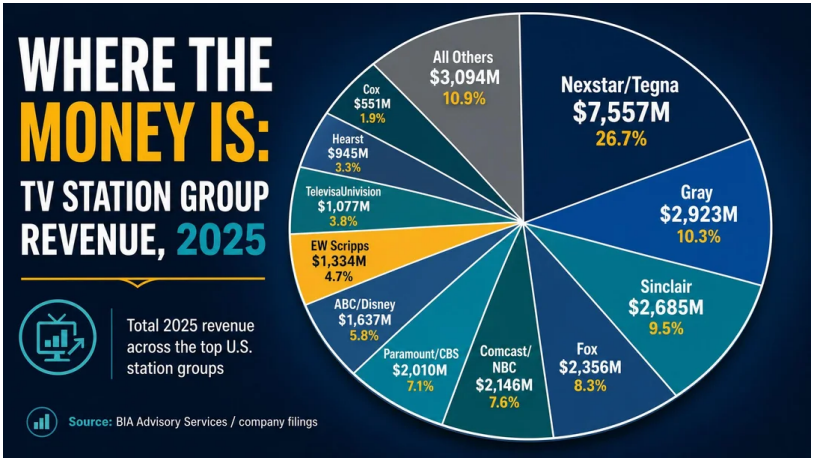
US COVID-19 cases on a spiral timeline



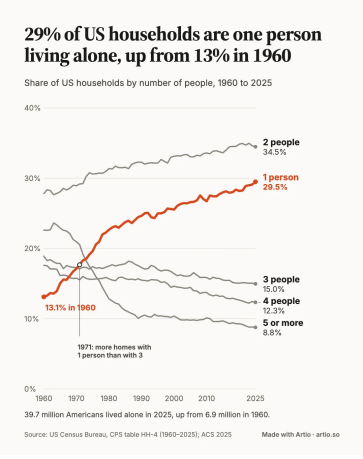
Building heights and the vertical scale



TV station group revenue, 2025



US household size over time



The week 1 assignment

How did total births change from month to month in 2014?

Redesign the supplied pie chart.

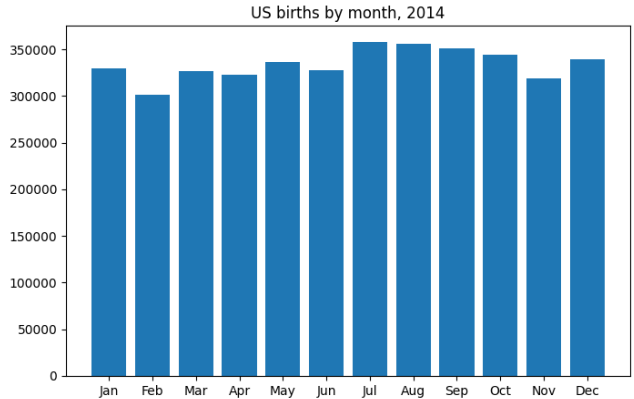
Keep all 12 months, the same totals and chronological order.

Add units and a note with source, year, aggregation and the month-length limitation.

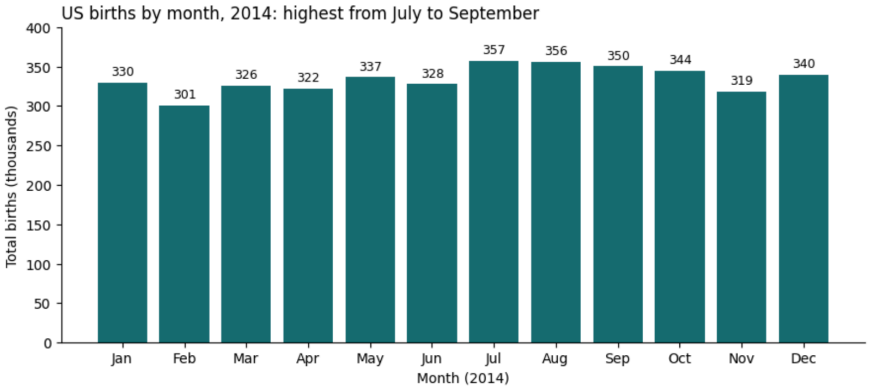
Check January and December against the table.

Optional extension: calculate and separately plot births per day.

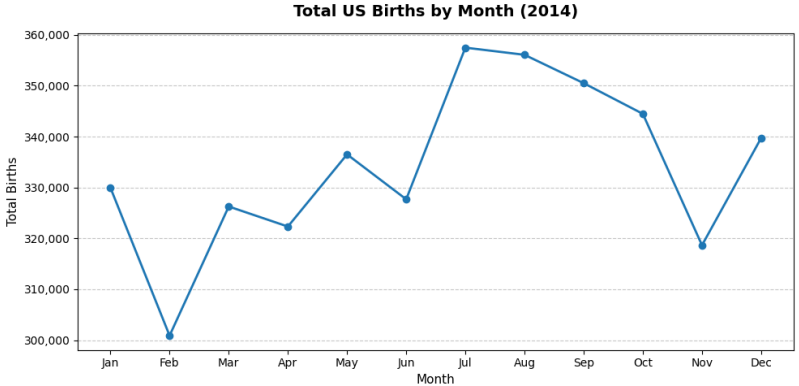
Submission A: chronological bars



Submission B: bars with a takeaway

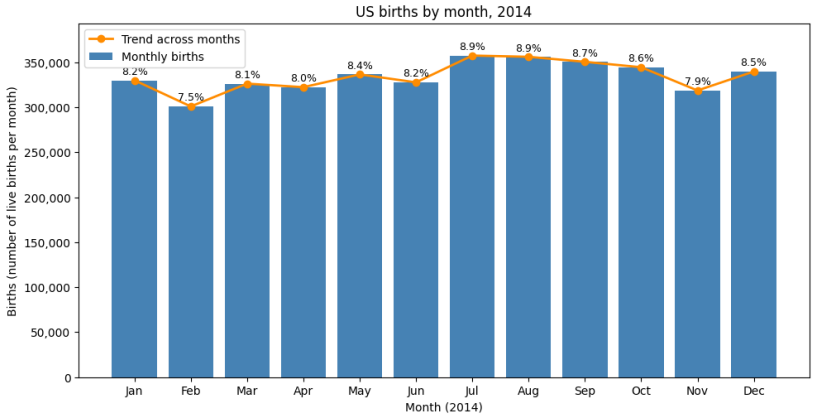


Submission C: a line with context



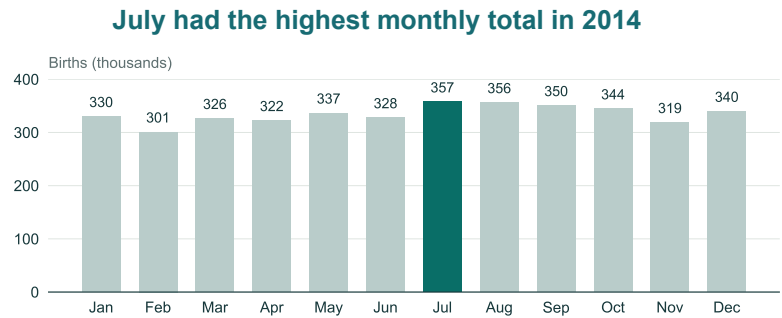
Note: Data shows monthly aggregated total counts for the year 2014.
Monthly totals reflect both birth frequency and the varying number of days per month (e.g., February vs. March).
Source: FiveThirtyEight / Social Security Administration administrative counts.

Submission D: bars, a line and annual shares



Note: Bars show total live births per calendar month, aggregated from daily counts; labels show each month's share of the 2014 total. Source: U.S. Social Security Administration (via FiveThirtyEight), 2014.

My attempt: monthly birth totals

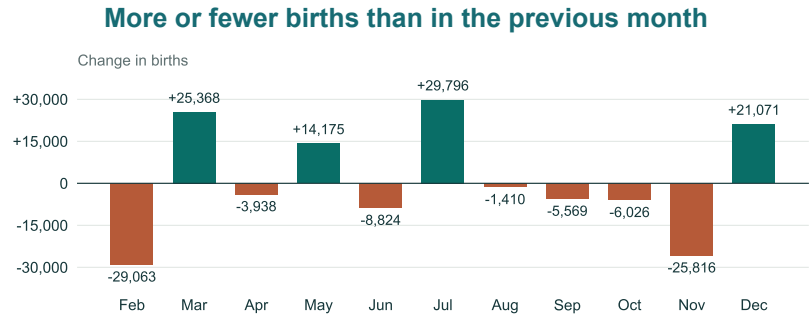


Notes: Monthly US birth totals in 2014, summed from 365 complete daily records. Labels are rounded to thousands. Totals are not adjusted for month length; the browser's daily view divides each total by calendar days.

Source: US Social Security Administration, via FiveThirtyEight.

Interactive chart (browser): monthly totals / births per day

My attempt: month-to-month changes



Notes: Each bar is the current month's US birth total minus the previous month's total, in births, for February-December 2014. January is omitted because December 2013 is outside the selected year. Negative values mean fewer births. Monthly totals include all recorded days and are not adjusted for month length.

Source: US Social Security Administration, via FiveThirtyEight.

Questions for today's lecture

1. What share of boot and sandal purchases cost $50 or less?
2. Do similarly warm months have similar boot-purchase shares?
3. Are boots cheaper out of season?

Question 1: what does $50 buy?

What share of boot and sandal purchases cost $50 or less?

If the average price exceeds $50, must most purchases cost more than $50?

What would you count, and what would you divide by?

How could one figure answer the question for several budgets?

$50 is a hypothetical budget per item. Each row is a donated purchase record.

Question 2: is temperature the whole story?

Do similarly warm months have similar boot-purchase shares?

Would you expect the same footwear choices in spring and autumn?

Should we compare boot counts, or the share of footwear purchases that are boots?

What information might a temperature-only graph leave out?

Purchase records matched to NOAA weather by shipping state, year and month.

Question 3: are boots cheaper out of season?

Compare summer boot prices with January.

Would weaker seasonal demand necessarily mean lower prices?

What sign would you expect for the summer price differences?

If an estimated difference is above or below zero, what else do we need to see?

Summer means June–August. January is the reference; year and shipping state are accounted for.

In-class exercise: redesign the chart

How do the seasonal purchasing patterns of Boots and Sandals differ?

Run the supplied chart cell: two pie charts of monthly purchases.

The data preparation is complete. Your task is to improve the visualisation.

You may change the chart type, layout, ordering, colours and labels.

Keep both categories, all twelve months and the same percentages.

Each category totals 100% across January–December, pooling the 2018–2022 purchase records.

What your redesigned figure must show

Make the two seasonal patterns easy to compare in calendar order.

Use clear scales, percentage units and category labels.

Add a note identifying the source, period, denominator and sample.

Check January for Boots and July for Sandals against the supplied table.

Explain two design choices that help answer the question.

15 minutes to redesign, then 5 minutes to compare figures. Chart type and layout are your choice.